WORKED SOLUTION

STEP II 2000, Question 4

Pure • Complex Numbers

STEP II 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q4

The key idea

The first two results say that when you multiply complex numbers, their arguments add. So an identity of the form "sum of arctangents $= \pi/4$" is really a statement about a product of Gaussian integers like $5 - \mathrm{i}$ and $1 + \mathrm{i}$: each factor $a + b\mathrm{i}$ with $a > 0$ contributes an angle $\arctan(b/a)$, and $\pi/4$ is the argument of anything on the line $y = x$. The work is to multiply out carefully, then to check that the angles really are equal and not just equal up to a multiple of 2π.

Solution

The product formula and de Moivre's theorem

Expanding and using $\mathrm{i}^2 = -1$,

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)(\cos\phi + \mathrm{i}\sin\phi) &= (\cos\theta\cos\phi - \sin\theta\sin\phi) + \mathrm{i}(\sin\theta\cos\phi + \cos\theta\sin\phi)\\ &= \cos(\theta+\phi) + \mathrm{i}\sin(\theta+\phi),\end{aligned}$$

by the compound-angle formulae for $\cos(A+B)$ and $\sin(A+B)$.

For the second result we use induction on n. When $n = 1$ both sides are $\cos\theta + \mathrm{i}\sin\theta$. Suppose the result holds for $n = k$. Then

$$\begin{aligned}(\cos\theta + \mathrm{i}\sin\theta)^{k+1} &= (\cos k\theta + \mathrm{i}\sin k\theta)(\cos\theta + \mathrm{i}\sin\theta)\\ &= \cos(k\theta+\theta) + \mathrm{i}\sin(k\theta+\theta),\end{aligned}$$

using the induction hypothesis and then the product formula with $\phi = k\theta$. So the result holds for $n = k+1$, and hence for every positive integer n.

It is convenient to write $\operatorname{cis}\theta$ for $\cos\theta + \mathrm{i}\sin\theta$. Then we have shown $\operatorname{cis}\theta\operatorname{cis}\phi = \operatorname{cis}(\theta+\phi)$ and $(\operatorname{cis}\theta)^n = \operatorname{cis} n\theta$.

A fact about angles

If $a > 0$ and b is real, then

$$a + b\mathrm{i} = \sqrt{a^2 + b^2}\,\mathrm{cis}(\arctan(b/a)),$$

because the angle $\alpha = \arctan(b/a)$ lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\cos\alpha > 0$, and $\tan\alpha = b/a$; these force $\cos\alpha = a/\sqrt{a^2+b^2}$ and $\sin\alpha = b/\sqrt{a^2+b^2}$.

Also, if $R\,\mathrm{cis}\,S = R'\,\mathrm{cis}\,T$ with $R, R' > 0$, then taking moduli gives $R = R'$, so $\cos S = \cos T$ and $\sin S = \sin T$, and therefore $S - T$ is a multiple of 2π.

The first identity

Let $\alpha = \arctan\frac{1}{5}$. Then $5 - \mathrm{i} = \sqrt{26}\,\mathrm{cis}(-\alpha)$ and $1 + \mathrm{i} = \sqrt{2}\,\mathrm{cis}\,\frac{\pi}{4}$, so by the results above

$$(5 - \mathrm{i})^2(1 + \mathrm{i}) = 26\sqrt{2}\,\mathrm{cis}(\tfrac{\pi}{4} - 2\alpha).$$

On the other hand, multiplying out directly,

$$(5 - \mathrm{i})^2 = 24 - 10\mathrm{i}, \qquad (24 - 10\mathrm{i})(1 + \mathrm{i}) = 24 + 24\mathrm{i} - 10\mathrm{i} + 10 = 34 + 14\mathrm{i},$$

and $34 + 14\mathrm{i} = \sqrt{1352}\,\mathrm{cis}\,\beta$ with $\beta = \arctan\frac{14}{34} = \arctan\frac{7}{17}$. Comparing, $\frac{\pi}{4} - 2\alpha$ and β differ by a multiple of 2π.

Ask yourself How do we know the multiple of 2π is zero?

We pin both angles down. Since $0 < \frac{1}{5} < 1$, we have $0 < \alpha < \frac{\pi}{4}$, so $-\frac{\pi}{4} < \frac{\pi}{4} - 2\alpha < \frac{\pi}{4}$. Also $0 < \beta < \frac{\pi}{2}$. Two angles in $(-\frac{\pi}{4}, \frac{\pi}{2})$ differ by less than 2π, so the multiple is zero and $\beta = \frac{\pi}{4} - 2\alpha$. That is,

$$\arctan\tfrac{7}{17} + 2\arctan\tfrac{1}{5} = \frac{\pi}{4},$$

as required.

The second identity

The same idea suggests looking at $(4 + \mathrm{i})^3(20 + \mathrm{i})(1985 + \mathrm{i})$, whose argument should be the left-hand side. Multiplying step by step:

$$\begin{aligned}(4 + \mathrm{i})^2 &= 15 + 8\mathrm{i},\\ (4 + \mathrm{i})^3 &= (15 + 8\mathrm{i})(4 + \mathrm{i}) = 52 + 47\mathrm{i},\\ (4 + \mathrm{i})^3(20 + \mathrm{i}) &= (52 + 47\mathrm{i})(20 + \mathrm{i}) = 993 + 992\mathrm{i}.\end{aligned}$$

This is already very close to the line $y = x$, and the last factor finishes the job:

$$\begin{aligned}(993 + 992\mathrm{i})(1985 + \mathrm{i}) &= (993 \times 1985 - 992) + (993 + 992 \times 1985)\mathrm{i}\\ &= 1970113 + 1970113\,\mathrm{i} = 1970113\sqrt{2}\,\mathrm{cis}\,\tfrac{\pi}{4}.\end{aligned}$$

(If you wondered where 1985 came from: asking for $(993+992\mathrm{i})(a+\mathrm{i})$ to have equal real and imaginary parts gives $993a - 992 = 992a + 993$, so $a = 1985$.)

Now let $p = \arctan\frac{1}{4}$, $q = \arctan\frac{1}{20}$ and $r = \arctan\frac{1}{1985}$. By the fact about angles, each of $4+\mathrm{i}$, $20+\mathrm{i}$, $1985+\mathrm{i}$ is a positive real multiple of $\operatorname{cis} p$, $\operatorname{cis} q$, $\operatorname{cis} r$ respectively, so the product is a positive real multiple of $\operatorname{cis}(3p+q+r)$. Comparing with the line above, $3p+q+r-\frac{\pi}{4}$ is a multiple of 2π.

Each of p, q, r lies in $(0, \frac{\pi}{4})$ because its tangent lies in $(0,1)$, so $0 < 3p+q+r < \frac{5\pi}{4}$. Hence $3p+q+r$ and $\frac{\pi}{4}$ differ by less than 2π, the multiple is zero, and

$$3\arctan\tfrac{1}{4} + \arctan\tfrac{1}{20} + \arctan\tfrac{1}{1985} = \frac{\pi}{4}.$$

Checks

Moduli must match: $|34+14\mathrm{i}|^2 = 1156+196 = 1352 = 26^2\times 2$, as it should. For the second product, $17^3\times 401\times(1985^2+1)$ should equal 2×1970113^2, and it does. Numerically, both left-hand sides evaluate to $0.785398163\ldots = \pi/4$ on a calculator; for instance $\arctan\frac{7}{17} \approx 0.390607$ and $\frac{\pi}{4} - 2\arctan\frac{1}{5} \approx 0.390607$.

Common slips

- Concluding equality of angles from equality of complex numbers without ruling out a difference of $2\pi k$. The range argument is a genuine part of the proof.
- Sign errors in $(5-\mathrm{i})^2$: it is $24-10\mathrm{i}$, so $5-\mathrm{i}$ contributes $-\arctan\frac{1}{5}$, which is why the arctangent appears with a plus sign on the other side.
- Arithmetic slips in the large product; check by the moduli, as above.

Beyond STEP

Identities like these are Machin-type formulae; John Machin's $\frac{\pi}{4} = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$, found the same way from $(5+\mathrm{i})^4(239-\mathrm{i})$, was used to compute π to a hundred places in 1706. The underlying structure is the arithmetic of the Gaussian integers $\mathbb{Z}[\mathrm{i}]$, where unique factorisation into Gaussian primes explains which such identities can exist. For the geometry of multiplication as rotation, read Needham, *Visual Complex Analysis*; for Gaussian integers, Davenport, *The Higher Arithmetic*.

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